

LLM-Based Anonymization of Psychiatric Medical Records

Context-aware de-identification via rephrasing, validated with tag-based, QA-based, and LLM-judge scoring.



878 patients · 7 model/prompt configurations · 3 validation methods

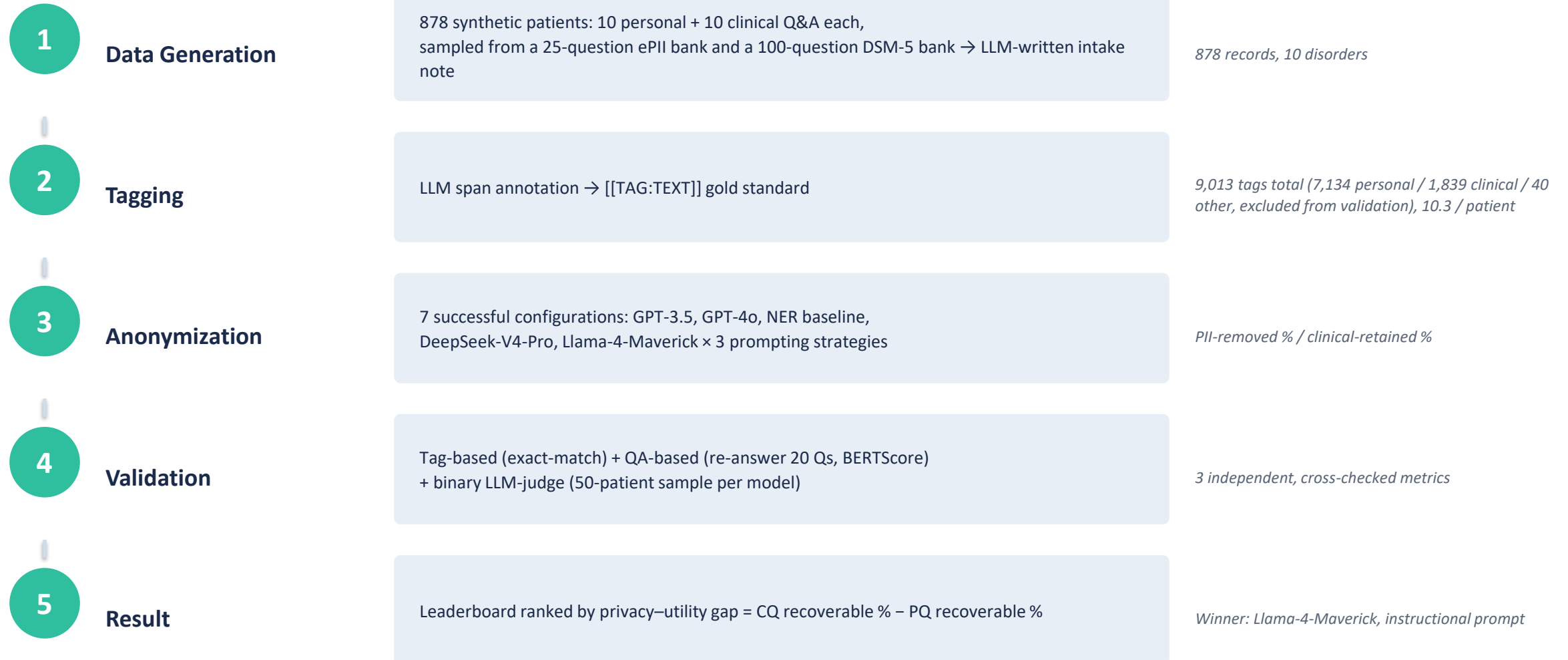
TEAM

Uriel Atzmon, ID – 209307172

Victoria Chuykina, ID - 321544512

Instructor: Alexander (Sasha) Apartsin, PhD

Five stages, end to end, all complete



What we compared, and how we scored it

MODELS COMPARED (ALL 878 PATIENTS)

- GPT-3.5-turbo, GPT-4o — instructional prompt
- NER baseline — entity masking, no rephrasing
- DeepSeek-V4-Pro, Llama-4-Maverick-17B — instructional prompt (Azure AI Foundry)
- Llama-4-Maverick — few-shot prompt variant
- Llama-4-Maverick — synthetic-identity prompt variant

EXCLUDED

gpt-5-mini: 657/878 rows failed (parameter-support errors) — left as documented future work, not part of headline comparisons.

THREE VALIDATION METHODS

T

Tag-based

Exact-span match: was each tagged PII span removed? Was each tagged clinical span preserved verbatim?

Q

QA-based (BERTScore)

Re-ask all 20 personal/clinical questions against the anonymized text only; score the new answer against the ground truth

J

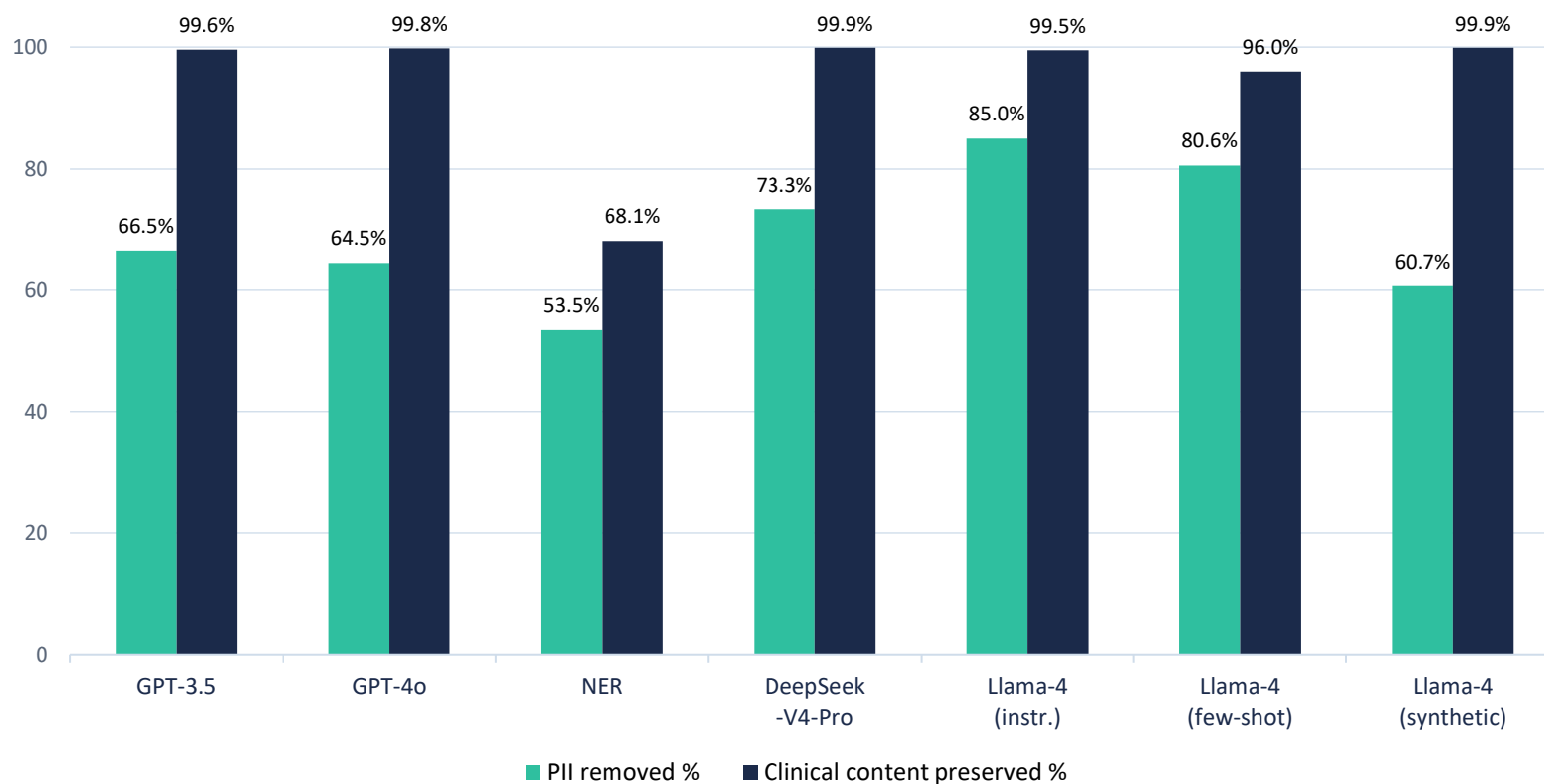
LLM-judge (binary)

Same 20 questions, but the LLM gives a direct yes/no: is this still recoverable from the note? 50-patient sample per model

Headline metric: privacy–utility gap = CQ recoverable % – PQ recoverable % (LLM-judge)

Tag-based validation, across all 7 configurations

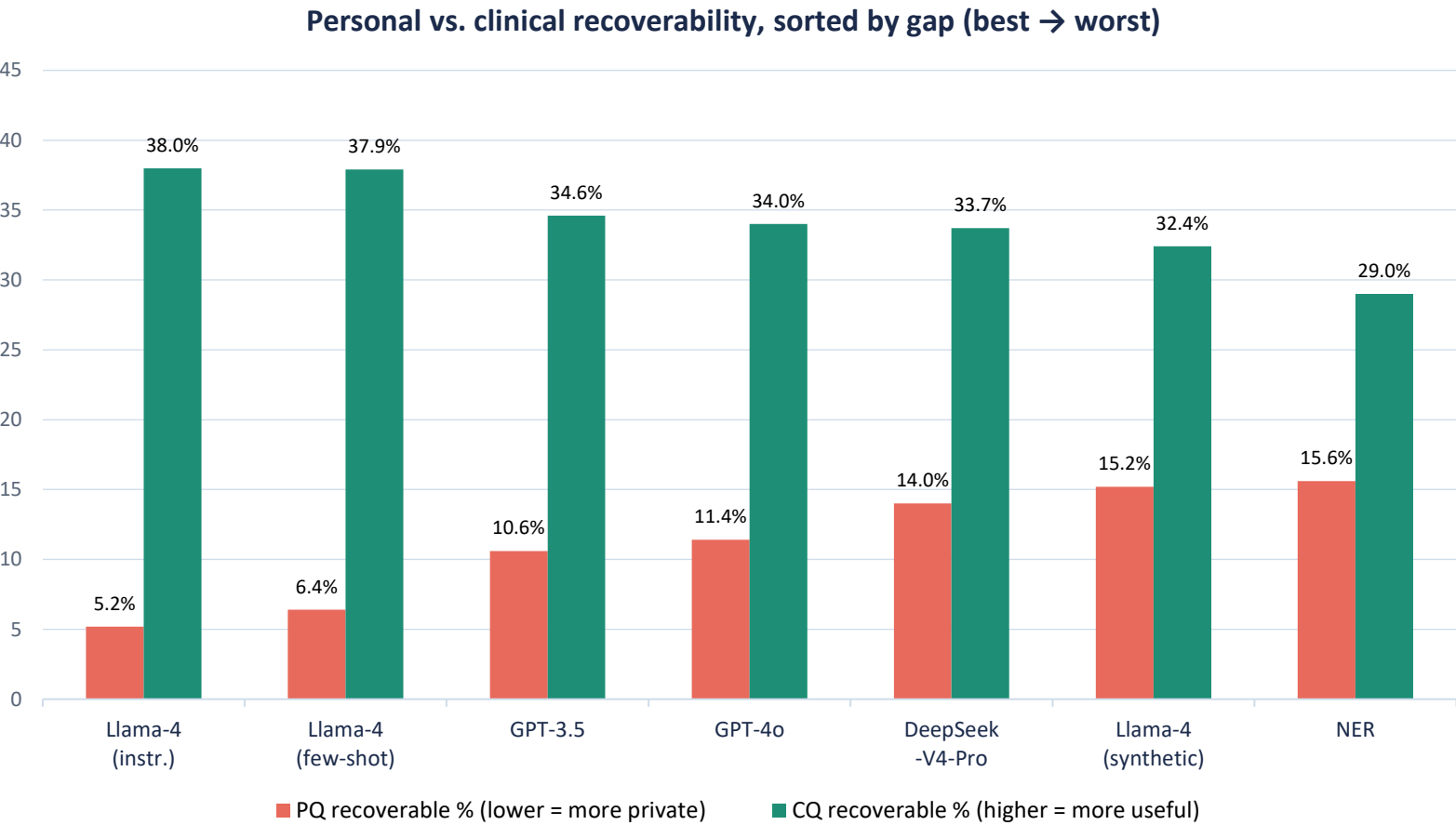
ePII removal vs. clinical retention, by model / prompt



READING THIS

- Llama-4-Maverick with the instructional prompt removes the most ePII (85.0%) while keeping clinical content nearly intact (99.5%)
- NER masking preserves the least clinical content by far (68.1%) — rephrasing beats blind masking
- Synthetic-identity prompting removes the least PII (60.7%) of the three Llama-4 variants — inventing a fake identity leaves more specific-sounding detail behind
- GPT-4o: 119/878 rows errored (API); DeepSeek-V4-Pro: 86/878 (rate limits) — both excluded from these percentages

QA-based validation: LLM-judge privacy-utility leaderboard



READING THIS

- Winner: Llama-4-Maverick, instructional prompt — gap of 32.8pp (PQ 5.2%, CQ 38.0%)
- Every configuration keeps personal-info recoverability low (5–16%) — privacy protection is solid everywhere
- CQ sits in a narrower 29–38% band — the judge applies a strict literal-fact standard to long clinical answers; treat tag-based clinical retention (95–99.9%) as the more reliable utility number
- Sample: 50 patients per model (cost/time), not the full 878
- Bias check: re-run with an independent judge (gpt-5-mini, no anonymizer overlap) — the winner holds, but “worst” does not (see Limitations)

One model, three prompts: does phrasing the instruction differently help?

Prompt variant	PII removed %	Clinical preserved %	PQ recoverable %	CQ recoverable %	Gap (pp)
Instructional (baseline)	85.0	99.5	5.2	38.0	32.8
Few-shot examples	80.6	96.0	6.4	37.9	31.5
Synthetic-identity	60.7	99.9	15.2	32.4	17.2

TAKEAWAY

- The plain instructional prompt wins on every single metric — few-shot examples and a synthetic-identity rewrite both underperform it
- Synthetic-identity prompting backfires on privacy specifically: asking the model to “invent a consistent fake identity” makes it keep more specific-sounding detail (60.7% PII removed vs. 85.0%), even though it still preserves clinical content well
- Prompt engineering alone did not beat the simple, direct instruction on this task — further gains likely require fine-tuning or a stronger base model, not prompt rewording

What we know is imperfect about these numbers

!	Dataset scale	878 synthetic patients — limited by generation API cost/time. Distributional balance across 10 disorders (68–106 patients each) is preserved. Could be scaled up further.
!	API error rows	GPT-4o: 119/878 (13.6%) and DeepSeek-V4-Pro: ~86/878 (9.8%) rows failed on transient Azure/OpenAI errors and are excluded from that model's percentages, not treated as anonymization failures.
!	gpt-5-mini incomplete	657/878 (75%) rows failed on parameter-support errors that were only partially self-healed at runtime. Excluded entirely from headline results; left as documented future work.
!	BERTScore floor effect	PQ and CQ BERTScore cluster tightly (0.886–0.911) regardless of real content overlap — short answers (e.g. “Not mentioned.”) score unexpectedly close to detailed ground truth under roberta-large embeddings. The binary LLM-judge is the more trustworthy metric for this reason.
!	LLM-judge sample size & self-preference	50-patient sample, not all 878. The judge (Llama-4-Maverick) is also 3 of the 7 configs it judges — checked with an independent judge (gpt-5-mini, no overlap): the winner holds, but NER is no longer clearly worst (Llama-synthetic is instead).

What we found, and where MindShield goes next

KEY FINDINGS

- Rephrasing beats entity masking on both axes: 99.5% clinical content preserved vs. 68.1% for NER, while also removing more PII (85.0% vs. NER's 53.5% — the lowest of all 7 configurations)
- Best configuration: Llama-4-Maverick with the plain instructional prompt — 85.0% PII removed, 32.8pp privacy–utility gap on LLM-judge
- Prompt engineering (few-shot, synthetic-identity) did not beat the simple instruction — synthetic-identity rephrasing actually leaked more PII
- Personal-info recoverability stays low (5–16%) across every model — privacy protection is robust regardless of which model anonymizes

FUTURE WORK

1

Scale the synthetic dataset further

2

Fix gpt-5-mini parameter support; re-run DeepSeek-V4-Pro's rate-limited rows

3

Run the LLM-judge on the full 878-patient set, not just samples

4

Clinician review of anonymized notes; multilingual extension; fine-tune a smaller self-hosted anonymizer on this benchmark